

Project Title: AI-Powered Skin Lesion Recognition System

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Abstract

SkinCare AI is a web-based skin cancer detection and classification system that utilizes deep learning techniques to analyze dermoscopic images of skin lesions. The system applies convolutional neural networks to classify uploaded images into eight different skin condition categories and provides prediction confidence scores. The primary goal of the project is to support early identification of potential skin abnormalities and encourage timely medical consultation. The application integrates a secure user interface, intelligent prediction logic, historical analysis tracking, report generation, and an AI-assisted chatbot, ensuring a responsible and informative approach to healthcare-oriented AI solutions.

Problem Statement

Skin cancer is one of the most prevalent cancers worldwide, and early diagnosis significantly improves treatment outcomes. However, access to dermatologists is limited in many regions, leading to delayed diagnosis and increased health risks. Additionally, the visual similarity between different skin conditions makes early identification difficult for non-experts. This project addresses these challenges by providing an accessible, AI-powered preliminary screening platform that assists users in identifying potential skin abnormalities and promotes medical awareness.

Objectives

Primary Objectives

- To develop an accurate deep learning model for skin lesion classification
- To provide a user-friendly web platform for image-based analysis
- To ensure secure user authentication and data management
- To generate structured reports that can assist medical consultations

Secondary Objectives

- To maintain analysis history for progress tracking
- To provide analytics and visual insights into prediction trends
- To integrate an AI chatbot for skin health guidance
- To ensure ethical and legally compliant AI result presentation

Module Details

1. User Authentication Module

- User registration with OTP-based email verification
- Secure login and logout
- Password reset functionality
- Admin authentication

2. Image Analysis Module

- Image upload and validation
- Image preprocessing and normalization
- Dual-model prediction system
- Confidence score calculation
- Result formatting with medical disclaimer

3. User Profile Module

- Profile information management
- Profile image upload
- User activity statistics

4. Analytics Dashboard Module

- Visualization of analysis trends
- Condition distribution charts
- User activity monitoring

5. History and Comparison Module

- Storage of past analysis results
- Side-by-side comparison of analyses
- PDF report generation

6. Admin Dashboard Module

- User management
- System usage monitoring
- Access control for administrative functions

7. DermaGenie AI Assistant Module

- AI-powered chatbot for skin health queries
- Context-aware responses
- Conversation history tracking

8. Email Notification Module

- Account verification emails
- Analysis notifications
- Password recovery emails

Dataset Information

The system is trained using dermoscopic image datasets including ISIC 2019 and a curated subset of the HAM10000 dataset. Images are preprocessed using resizing, normalization, and data augmentation techniques. The dataset supports classification into eight skin condition categories.

Minimum Requirements

- Processor: Intel Core i5 or equivalent
- RAM: 8 GB
- Storage: 10 GB free space
- Internet connection

Software Requirements

- Operating System: Windows / Linux
- Programming Language: Python 3.10+
- Web Framework: Django
- ML Frameworks: TensorFlow, Keras
- Database: SQLite (development)
- Frontend: HTML, CSS, JavaScript
- Development Tools: VS Code / PyCharm

Conclusion

SkinCare AI demonstrates the effective use of artificial intelligence in healthcare applications by combining deep learning models with a secure and user-friendly web platform. The project highlights the potential of AI-assisted preliminary screening systems while maintaining ethical standards and encouraging professional medical consultation.

Counter Signed

Project Guide

Mentor

Project Coordinator

Head of the Department